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A Comparison of Traditional Natural Language Processing Methods for Emotion Recognition using Quantum Computing

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***Abstract*—Emotion Recognition plays an important role in recognizing human behavior. Human behavior may be represented in text, speech, and facial expressions. In traditional form, natural language processing known as NLP is used to extract features to evaluate emotions. With the evolution of Quantum Computing, Quantum Machine learning often called QML is one of the research areas used to evaluate emotions. This paper is designed to compare traditional ML models with QML models to analyze the sentiments and emotions of multimodal data. Here, NLP techniques are used for preprocessing. A hybrid model is built by combining NLP and QML. The preprocessed data are transmitted into quantum circuits for computation, and its results are compared with conventional machine learning models. This evaluation identifies QML as a promising future as it is useful for large and complex data sets with high dimensions. This study highlights the potential of QML in NLP and outlines the challenges such as encoding and hardware limitations.**

***Index Terms*—Natural language processing, quantum computing, quantum machine learning, machine learning**

I. Introduction

With its ability to comprehend, interpret, and produce human language, natural language processing has emerged as a crucial element of artificial intelligence. Sentiment analysis and emotion recognition are two important uses of natural language processing (NLP) that aid in business and public opinion gathering by examining text data like tweets, reviews, and customer feedback. Conventional NLP approaches process and analyze text data using machine learning and deep learning techniques. These techniques call for many procedures, such as extraction of features, stop word elimination, and tokenization. This method is very helpful for recognizing emotions and for recognizing important words like ”love” and ”hate,” while downplaying words like ”the” and ”of”. Word embeddings, like Word2Vec and GloVe, which encode words as dense vectors in a multi-dimensional space, are another popular method in classical natural language processing. Models are able to comprehend that words like ”happy” and ”joyful” have closely linked meanings because these embeddings reflect the semantic relationships between words. A model trained on word embeddings, for instance, may identify that the sentence ”I am happy” transmits a positive attitude and ”I am sad” conveys a negative sentiment in a sentiment analysis test[4][5][6]. The traditional NLP techniques are successful but face challenges with increasing data complexity and volume. For example, processing a large number of tweets during an international event demands significant computational power. Quantum Machine Learning (QML) offers an alternative, leveraging qubits to process high-dimensional data more efficiently[7]. Grover’s search technique, for instance, is quadratically faster than traditional algorithms at searching unsorted databases, which may be useful for tasks like emotion analysis or keyword extraction on big datasets. Recent studies have investigated the possibility of QML in NLP tasks. for example, shows that machine learning models could perform better in classification tasks when feature spaces are quantum-enhanced [8][9]. To capture more intricate word associations, they employed quantum circuits to map classical data into a high-dimensional quantum feature space. This method might be beneficial for tasks like emotion detection, where complex models are needed to effectively understand emotional cues due to the complexities of human language. QML’s use in NLP is still in its infancy, though, and there are still several obstacles to overcome. High-dimensional NLP data encoding into quantum circuits is one of the main obstacles [10][11].

II. Methodology

A. Data Collection And Preprocessing

In this paper, The dataset is taken from Kaggle is a supervised dataset consisting of 22,000 tweets on the football World Cup 2022 and emotions are analyzed using text2emotion[12]. To ensure this dataset can be effectively applied with both classical machine learning and the Quantum machine learning models, It undertook a comprehensive preprocessing phase that included several essential steps. One of the most crucial steps in this preprocessing phase is known as tokenization. This specific process involves dividing phrases and sentences into smaller, distinct units called tokens [14]. By doing this, we simplify the process of converting text data into a numerical

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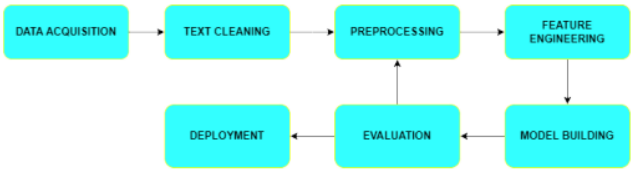


Fig. 1. Traditional pipeline for the step-by-step procedure in Classical NLP Methods

values, which is extremely important for any subsequent analysis intend to perform. Another vital step in preprocessing work is known as stop word removal [15]. In the preprocessing stage, data cleaning is important as it removes stopwords that are not needed in the context. Identify the special characters, nonalphanumeric characters, emoticons, URLs, and any digits that appear are removed from tweets[16]. It makes data in a well-organized form. After this lemmatization and stemming are done. Then convert this data into lowercase. This lowercase ensures uniformity throughout this dataset.

B. Feature Extraction And Traditional Methods

The study focused on establishing a benchmark based on the traditional natural language processing pipeline, which consists of several key steps i.e. necessary to analyze and understand textual data in Figure.1.

This data is ready for feature extraction. To extract features, Natural Language Processing techniques such as count vectorization (CV) transform the text into vectors based on the frequency of each word in the entire dataset, Term Frequency Inverse Document Frequency (TF-IDF) gives importance of a word within a specific document while considering its overall relevance throughout a collection of documents[16]. It assists in emphasizing terms that are important in specific documents while minimizing words that appear often in multiple texts. and word embeddings such as Word2Vec and Glove models are utilized to effectively represent the relationships between words in a vector space[17][18]. while transitioning to quantum machine learning models, the input lengths are consistent to guarantee this quantum encoding is facilitated[19].

This is achieved through a couple of methods. one of the methods is by statistical metric evaluation and the other approach is TF-IDF [20] gives importance of a word within a specific document while considering its overall relevance throughout a collection of documents. The values for TF-IDF are calculated using formulae described below:

$$tfidf(i,k,K) = tf(i,k).idf(i,k,K)$$

where Term Frequency (tf) is given by:

$$tf(i,k) = \frac{f_{i,k}}{\sum_{i' \in k} f_{i',k}}$$

and Inverse Document Frequency (idf) is given by:

$$idf(i,k,K) = \log \frac{N}{|\{k \in K : i \in k\}|}$$

It assists in emphasizing terms that are important in specific documents while minimizing words that appear often in multiple texts.

C. Model of Quantum Machine Learning

To achieve quantum embedding in quantum machine learning, a technique known as angle encoding was utilized. This method converts numerical input values into the rotation angles of quantum gates. By employing this embedding technique, To efficiently encode high-dimensional data [25][26]. This is made possible by leveraging the exponential growth of Hilbert space, which allows us to represent complex data structures in a more manageable way.

Figure 2 shows the architecture of Quantum Circuits for emotion recognition in the Quantum Model, a simulated quantum device was used, and it consisted of qubits that were equal in number to the padded length of each tweet. The quantum circuit was made up of several essential components:

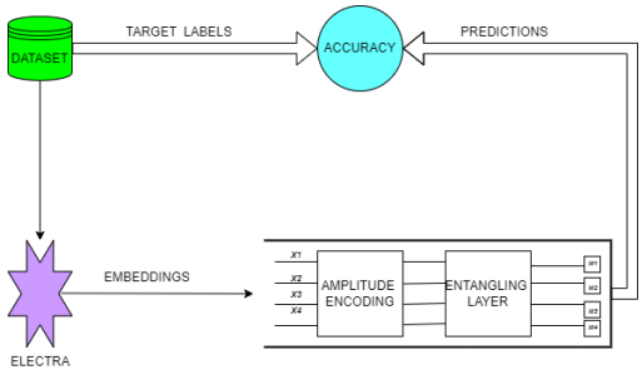


Fig. 2. The Architecture of Quantum Circuit for Emotion Recognition. The schematic diagram defines quantum circuits that encode and process data for emotion classification.

Angle Embedding: This part of the circuit was responsible for embedding the input data into rotation angles. This preparation was crucial in establishing the initial quantum states needed for processing the information [27].

$$|\psi_{\text{embedded}}\rangle = \prod_{i=1}^{n} R_Y(x_i)|0\rangle$$

Parameterized Gates: Specifically, the RX, RY, and RZ gates were included. These gates are designed for each qubit and come with trainable rotation capabilities. This feature allows the circuit to incorporate learnable parameters, making it adaptable and capable of optimizing its performance [28].

$$|\psi_{\text{param}}\rangle = \prod_{i=1}^{n} R_X(\theta_{i,1}) R_Y(\theta_{i,2}) R_Z(\theta_{i,3}) |\psi_{\text{embedded}}\rangle \quad (1)$$

Entanglement Layers: To enhance the representation of features, entanglement was selectively added to the circuit. This use of quantum entanglement allows for better capture of interactions between different features [29]. By doing so, it enables a deeper understanding of how different data elements

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relate to each other, which is beneficial to improve the model's accuracy and effectiveness.

$$|\psi_{\text{entangled}}\rangle = \prod_{i=1}^{n-1} \text{CNOT}(i, i+1)|\psi_{\text{param}}\rangle \quad (2)$$

Measurement: The expectation value of the Pauli-Z operator was measured for classification [30].

$$\langle Z\rangle_i = \langle\psi_{\text{final}}|Z_i|\psi_{\text{final}}\rangle \quad (3)$$

Algorithm for Quantum Computing Methodology:

1. ***Dataset***:

Football Tweets 2022 from Kaggle.

2. ***Data Preprocessing*****:**

(i)Data Cleaning using NLP Techniques.

(ii)Annotation Of Tweets.

3.***Build The Vocabulary***:

(i)Find Maximum length of pre-processed tweets.

(ii)Create word to index mapping.

(iii)Convert Tweets into numerical values.

(iv)Padding With Zeros.

4.Build a Quantum Circuit**:**

(i)Define Qubits based on length of tokenized text.

(ii)Perform Quantum Embedding by converting numerical text features into quantum rotational angles.

(iii)Intoduce trainable quantum gates for feature transformation.

(iv)Apply Entanglement using C-NOT gates to establish correlation.

(v)Perform Cost Function & Optimization on the transformed features.

5.Execute the quantum circuit on a quantum simulator:

(i)Measure Pauli-Z Expectation values for output classification.

(ii)Process measurement results and classify emotions.

6.Evaluate the model on test dataset.

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III. Results And Discussions

The proposed algorithm was executed in the Google Colab Python environment and the dataset was taken from Football Tweets-2022 Kaggle repository.

The data visualization provides an essential framework for illustrating the results of study comparing quantum machine learning (QML) with classical natural language processing (NLP) methods in the context of tweet sentiment analysis and Emotion Recognition. This comparison is crucial in understanding the performance, advantages, and challenges of each technique.

Formulae to calculate Accuracy, Precision, and Recall:

$$Accuracy = \frac{TN + TP}{TN + FN + TP + FP}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

A. Discussion Of Results

In Emotion Recognition, performance metrics were evaluated for both classical and quantum models. The classical models achieved notable results, where as Support Vector Machine (SVM) demonstrating an accuracy of 84.3%, while the Random Forest (RF) model performed slightly better at 85.7%.

In contrast, the quantum models exhibited lower accuracy, with the single-layer quantum circuit achieving 78.5% and the two-layer circuit improving slightly to 80.2%. Precision and recall values for the quantum models also reflected these trends, highlighting the quantum models challenges when compared to their classical counterparts. This analysis underscores the potential of quantum computing in sentiment analysis,while also pointing to its current limitations. The following table shows the performance metrics comparison which illustrates how the proposed algorithm's accuracy, precision, and recall values differed from existing SVM and RF methods.

The demonstration of quantum computing [26] highlights the feasibility of quantum processors outperforming classical computers on specific computational tasks, suggesting potential advantages for NLP applications.

Quantum algorithms for supervised and unsupervised learning, such as those proposed by Lloyd et al. [27], demonstrate how quantum feature spaces could enhance NLP tasks by efficiently encoding high-dimensional data.

B. Visualization Of Results

Model	Accuracy	Precision	Recall
Classical (SVM)	84.3%	82.7%	83.1%
Classical (RF)	85.7%	84.2%	84.5%
Quantum (1 layer)	78.5%	77.3%	76.9%
Quantum (2 layers)	80.2%	78.9%	78.5%

TABLE I
THE PERFORMANCE METRICS COMPARISON ON CLASSICAL (VS) QUANTUM METHODS

Below figures 3, 4, and 5 are the key visual components that will be incorporated to effectively convey the findings: Figure 3 shows the accuracy, precision, and recall results of the proposed algorithm. Figure 4 illustrates confusion matrix of the proposed algorithm. where as Figure 5 illustrates the comparative results between the proposed algorithm and existing SVM, RF algorithms

From the below confusion matrix the observations are:

Surprise has the highest percentage of accurately classified categories (1995 correct predictions).There is substantial confusion between the emotions of anger and surprise (252 misclassified).There is some confusion between the emotions of fear and surprise (135 misclassified).

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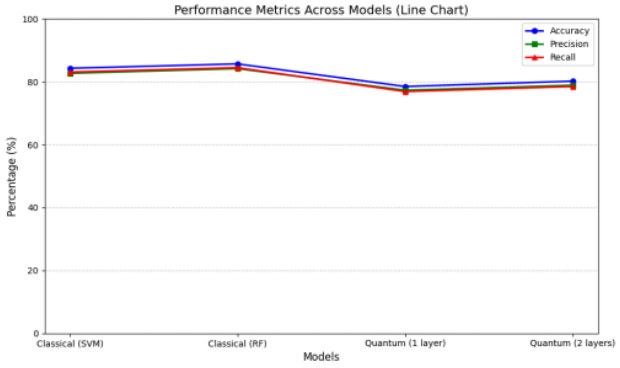


Fig. 3. The Performance Metrics Comparison: A line chart showing accuracy, precision, and recall for classical and quantum methods.

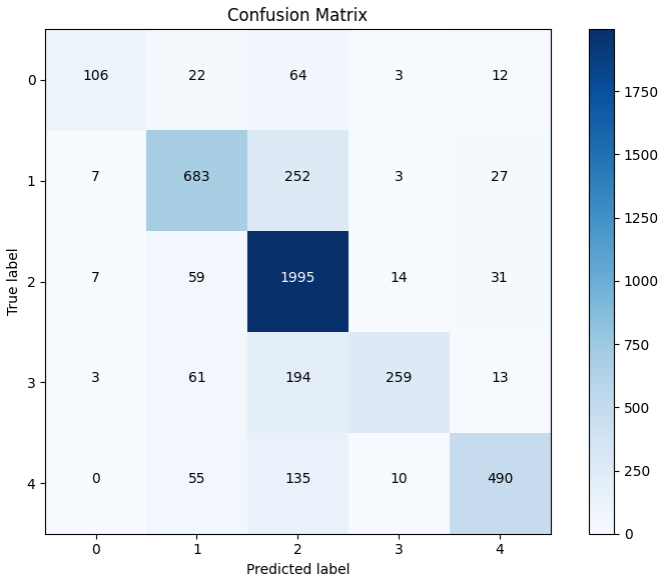


Fig. 4. Confusion Matrix illustarting the quantum model performance on test data in classfying emotions.

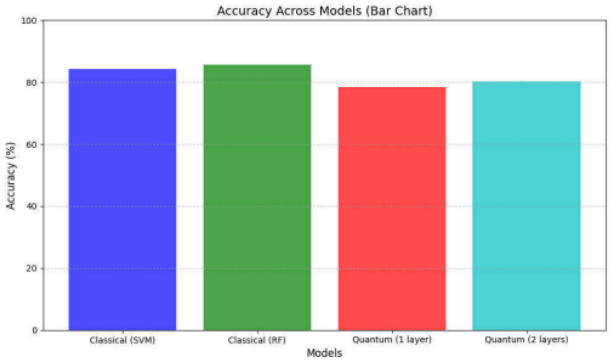


Fig. 5. Accuracy Comparison Across Models: A bar chart comparing the classification of the accuracy of SVM, Random Forest, and Quantum Models.

C. Observations

Accuracy Gap:

The accuracy of classical models was consistently higher than that of QML models. The greater performance of classical algorithms can be attributed to their maturity as well as effective feature extraction methods like word embeddings [5][17]. However, the goal of this study is to illustrate the special benefits and difficulties of QML, not to outperform classical models. Although hardware and optimization technique restrictions cause QML to perform poorly at the moment, its potential to explore new feature spaces makes it an attractive area for future research [8][9]. Scalability: Although QML struggled with scalability, it demonstrated promise in managing small datasets. The need to encode high-dimensional NLP data into quantum circuits is the source of this restriction. exchanges [10][12].

Interpretability:

When it came to feature exploration, quantum models showed special advantages. Complex relationships in the data can be naturally captured by QML circuits, in contrast to traditional models that rely on pre-defined kernels or activation functions. This implies that quantum circuits may be able to reveal relationships and feature importance that may be hidden in classical models [18][27].

D. Challenges Of QML

Quantum Noise and Decoherence

The stability and precision of quantum circuits are affected by ambient noise and decoherence. These effects are particularly noticeable in noisy intermediate-scale quantum (NISQ) devices, which are now the most popular experimental hardware. To tackle these problems, error mitigation strategies including quantum error correction and zero-noise extrapolation are being developed [12][28].

Embedding High-Dimensional NLP Data

One major obstacle is still converting text data into a format that can be processed by quantum computing. Although it works well with low-dimensional data, angle embedding has scalability issues when used for high-dimensional NLP tasks. Although they present issues for NISQ devices, advanced encoding techniques like amplitude encoding and quantum feature maps offer potential answers but call for extra qubits and computational power [13][19].

IV. Conclusion

The results of research reveal that quantum machine learning, with its superior processing capabilities, significantly improve the accuracy of various NLP applications which makes efficient. Given the vast amounts of unstructured data available today, the ability to analyze and interpret this data swiftly and accurately is of paramount importance. As organizations increasingly rely on real-time data analysis to inform decision-making, tools grounded in QML could become indispensable. The implications for businesses, governments, and other entities that rely on sentiment analysis and emotion recognition are substantial, as more understanding of public sentiment

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can drive more effective communication strategies, marketing campaigns, and policy decisions.

As complex datasets are generated Quantum architectures play a great role in performing computations. So future research focuses on the architecture of quantum machines and algorithms are developed for Quantum Machine Learning. So more frameworks to be developed to increase the performance of QML

V. Acknowledgement

I would like to sincerely thank everyone who helped me with the production of this research on sentiment analysis and emotion recognition using quantum machine learning. Above all, I would want to express my gratitude to the academic advisor, whose advice and perceptive criticism greatly influenced the direction of my research. I was motivated to pursue excellence and to push the limits of knowledge by the support.

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